Bidipta Roy

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Education

Boston University ~ *Bachelor of Science in Computer Science* — Boston, MA

Sep. 2023-May 2027

- Relevant coursework: Data Structures and OOP, Software Engineering, Probability, Discrete Math, Computer Systems, Linear Algebra, Algorithms, Databases

Technical Skills

Languages: Python, Java, JavaScript, C, HTML/CSS, x86 Assembly
Technology: React, Node.js, Express, Flask, Tailwind, REST APIs, SQL
Developer Tools: Git, VScode, MongoDB Atlas, Vercel, Figma, Docker

Employment

Swim Instructor - Long Island Swim School — Garden City, NY

Sep 2021 - Aug 2023

• Taught group lessons focusing on stroke fundamentals, technique improvement, and water safety for ages 4–12 using personalized training plans; taught 100+ students

Swim Coach - SWIMtastic swim school — Hempstead, NY

May 2025 - Aug 2025

• Coached youth and adult swimmers in 1:1 and small groups, creating individualized plans to improve stroke efficiency; managed roster of students flowing throughout the week for all instructors

Projects

Nogi BJJ Trainer | *React, Node.js, Express, MongoDB, Tailwind CSS* — November 2025

- Built a full-stack web application with 19+ Jiu Jitsu technique modules across 6 categories, implementing RESTful API endpoints, and real-time quiz polling with live vote aggregation
- Integrated MongoDB cloud database to store profile info and comments, and secure API routes, deploying frontend on Vercel with CI/CD pipeline and backend on Node.js server
- Designed responsive UI with Tailwind and React Hooks, implementing CRUD operations for comment system and session storage

Spotter | *Python, Flask, Javascript, HTML/CSS, MongoDB, OpenAI API, JWT* — December 2025

- Implemented Flask backend with RESTful endpoints for challenges, workout logs, and activity feeds using a lightweight JSON store.
- Collaboratively built a responsive multi-page frontend using JavaScript, HTML, and CSS to help users track and see their physical activity progress
- Integrated OpenAI's API to generate personalized one-meal recipes based on user's preferences using structured JSON responses.

EcoRoute | *React with Vite, Tailwind CSS, Mapbox GL maps API, Node/Express.js, SQLite* — February 2026

- Maps and visualizes trash/recycle Big Belly Bins across BU's three campuses using real sensor data from 84 Bins with plans to implement dynamic data
- Developed a BU sustainability dashboard using a hotspot algorithm to identify overflowing bins, optimize collection times, and improve placement based on foot traffic.
- Cleaned and processed raw CSV data from City of Boston's open data portal and BU's Big belly sensor network into a SQLite database

Leadership and Community Involvement

Boston University Jiu Jitsu Club — Jan 2025-Present

• Practice, teach, and develop the martial art of brazilian jiu jitsu and conditioning

NY Monarch Jiu Jitsu Academy — May 2025-Present

• Train four days a week to develop advanced grappling technique and discipline

BU Hack4Impact Club — Sep 2025-Present

• Participate in workshops and collaborative web projects using JS, HTML/CSS, React